MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $(2,1)\cdot(3,-4)$ equals
   - A. 2
   - B. 10
   - C. $(6,-4)$
   - D. -2
2. The norm of $(-3,4)$ equals
   - A. 7
   - B. 25
   - C. 5
   - D. 1
3. Two vectors with zero product are
   - A. equal
   - B. collinear
   - C. opposite
   - D. orthogonal
4. Norms 2 and 3, angle $60°$: product
   - A. 6
   - B. 3
   - C. $\sqrt{3}$
   - D. 0
5. $\|u-v\|^2$ equals
   - A. $\|u\|^2-\|v\|^2$
   - B. $\|u\|^2+\|v\|^2$
   - C. $\|u\|^2-2u\cdot v+\|v\|^2$
   - D. $\|u\|^2+2u\cdot v+\|v\|^2$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. An obtuse angle between nonzero vectors gives a product that is
   - A. positive
   - B. zero
   - C. equal to 1
   - D. negative
2. In an orthonormal basis, $u\cdot i$ is
   - A. the x-coordinate of u
   - B. the norm of u

C. the y-coordinate of u
D. always zero

3. The locus $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for $A \neq B$ is
   A. the perpendicular bisector
   B. the line AB
   C. the circle with diameter AB
   D. the midpoint alone
4. If $u \cdot v = 3$, $(2u) \cdot (-v)$ equals
   A. -3
   B. -6
   C. 6
   D. 9
5. For $AB = AC = 2$ and a right angle at A, BC equals
   A. 4
   B. 2
   C. 8
   D. $2\sqrt{2}$

True or false? Justify.

1. The scalar product of two vectors is always positive.
2. The zero vector is orthogonal to every vector.
3. $u \cdot v = 0$ implies that one of the vectors is zero.

> **Key point · Self-assessment.** I distinguish the norm from its square; I choose a suitable formula; I check the frame; I write an equivalence for a locus.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.